

使用wsl2训练智能小车循线模型

安装wsl

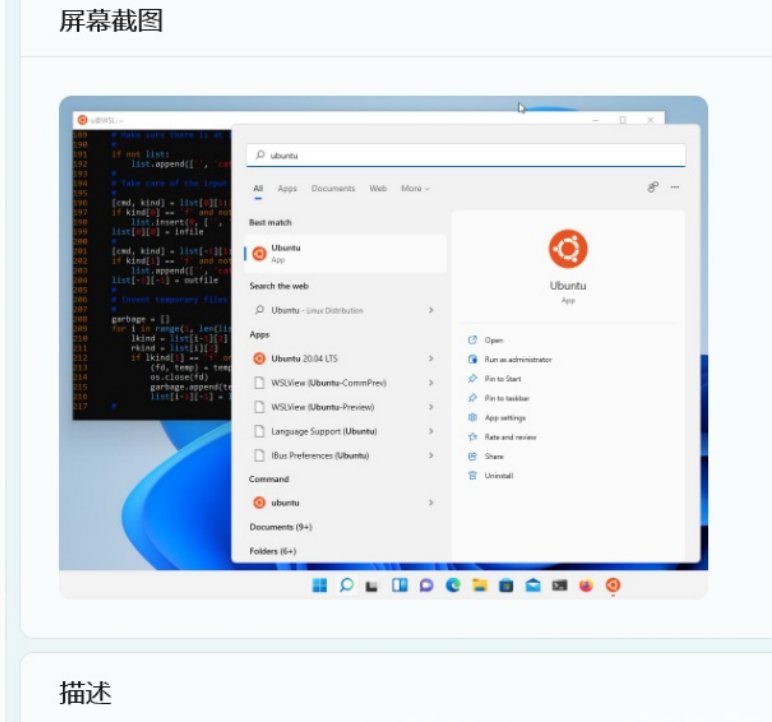
官方文档[设置wsl开发环境](#)

安装Windows终端是可选项，强烈建议安装，使用体验会上升，以下操作都是使用Windows终端进行的。

安装完wsl后可以在Microsoft Store安装一个Ubuntu发行版，这里使用的是Ubuntu20.04.4



屏幕截图



可选：可以在Windows终端设置默认配置

	Windows PowerShell	Ctrl+Shift+1
	命令提示符	Ctrl+Shift+2
	Azure Cloud Shell	Ctrl+Shift+3
	Ubuntu 20.04.4 LTS	Ctrl+Shift+4
	设置	Ctrl+,
	命令面板	Ctrl+Shift+P
	关于	

默认配置文件

单击 '+' 图标或通过键入新的选项卡键绑定定时打开的配置文件。

之后使用Windows终端新建一个标签页就是Ubuntu20.04.4LTS的终端了。

若未安装Windows终端可在开始菜单找到刚刚安装的Ubuntu，或在搜索栏直接搜索



开始配置caffe框架(cpu)

参考教程[ubuntu20.04安装caffe和配置anaconda](#)

配置好后在终端输入caffe命令，应该会出现下面的输出(反正我的是这样的)

```
xianyu@LAPTOP-V9JP5BP5: ~/ | x x x
Code\bin
(base) xianyu@LAPTOP-V9JP5BP5:~$ conda activate caffe
(caffe) xianyu@LAPTOP-V9JP5BP5:~$ caffe
caffe: command line brew
usage: caffe <command> <args>

commands:
  train          train or finetune a model
  test           score a model
  device_query   show GPU diagnostic information
  time          benchmark model execution time

Flags from /build/caffe-n38wHx/caffe-1.0.0+git20180821.99bd997/tools/caffe.cpp:
-gpu (Optional; run in GPU mode on given device IDs separated by ','. Use
'-gpu all' to run on all available GPUs. The effective training batch
size is multiplied by the number of devices.) type: string default: ""
-iterations (The number of iterations to run.) type: int32 default: 50
-level (Optional; network level.) type: int32 default: 0
-model (The model definition protocol buffer text file.) type: string
default: ""
-phase (Optional; network phase (TRAIN or TEST). Only used for 'time'.)
type: string default: ""
-sighup_effect (Optional; action to take when a SIGHUP signal is received:
snapshot, stop or none.) type: string default: "snapshot"
-sigint_effect (Optional; action to take when a SIGINT signal is received:
```

智能小车算法开发

参考官方文档[智能小车算法开发](#)

主要根据官方文档做，这里给一些踩过的坑

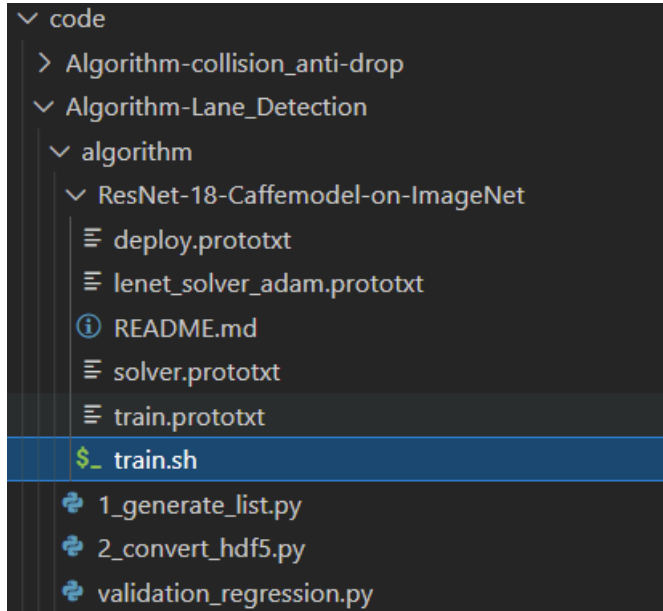
- 如果ATool用不了的话可以换一台电脑，用老师给的虚拟机标注完数据，再拷贝到wsl里面
- 如果运行脚本文件1_generate_list.py和2_convert_hdf5.py出错而且怎么也解决不了，也可以用虚拟机运行完后将生成的hdf5文件夹下的内容拷贝到wsl

总的来说就是其他的步骤都可以用虚拟机完成，只是训练模型使用wsl会快很多。从下面两张图可以看出，从1:08开始到11:56就完成了5000次迭代。速度相较于虚拟机还是快了许多的。

```
I0707 01:08:38.337234 468 layer_factory.hpp:77] Creating layer data
```

```
I0707 11:56:55.917728 487 solver.cpp:327] Iteration 5000, loss = 0.00021212
I0707 11:56:55.917771 487 solver.cpp:332] Optimization Done.
I0707 11:56:55.917785 487 caffe.cpp:250] Optimization Done.
```

最重要的是训练之前需要更改一些文件里的路径，否则会报错



这个目录下的

train.sh, train.prototxt, solver.prototxt, lenet_solver_adam.prototxt 都要改

```
code > Algorithm-Lane_Detection > algorithm > ResNet-18-Caffemodel-on-ImageNet > $._ train.sh
1 # export CAFFE_ROOT=$PWD
2 # export PATH=$CAFFE_ROOT/build/tools:$PATH
3 # export PYTHONPATH=$CAFFE_ROOT/python:$PYTHONPATH
4 # export LD_LIBRARY_PATH=/home/hanxb/anaconda3/envs/caffe-ssd/lib:$LD_LIBRARY_PATH
5
6 caffe train \
7     -solver lenet_solver_adam.prototxt \
8     -weights /home/xianyu/programming/atlas200dk/code/tools/ATool/model/road_following_model.caffemodel \
9     # -gpu 0
10
```

这里batch_size设置的是32，之前用的是128，训练的时候被kill掉了，根据电脑性能量力而行吧

```
code > Algorithm-Lane_Detection > algorithm > ResNet-18-Caffemodel-on-ImageNet > train.prototxt
1 name: "ResNet-18"
2 layer {
3     name: "data"
4     type: "HDF5Data"
5     top: "data"
6     top: "label"
7     include {
8         phase: TRAIN
9     }
10    hdf5_data_param {
11        source: "/home/xianyu/programming/atlas200dk/code/tools/ATool/hdf5/train_h5.txt"
12        batch_size: 32
13    }
14 }
```

```

15 layer {
16     name: "data"
17     type: "HDF5Data"
18     top: "data"
19     top: "label"
20     include {
21         phase: TEST
22     }
23     hdf5_data_param {
24         source: "/home/xianyu/programming/atlas200dk/code/tools/ATool/hdf5/test_h5.txt"
25         batch_size: 32
26     }
27 }

```

```

code > Algorithm-Lane_Detection > algorithm > ResNet-18-Caffemodel-on-ImageNet > solver.prototxt
1 net: "/home/xianyu/programming/atlas200dk/code/Algorithm-Lane_Detection/algorithm/ResNet-18-Caffemodel-on-ImageNet/train.prototxt"
2 test_iter: 8
3 test_interval: 300
4 base_lr: 1e-3
5 display: 10
6 lr_policy: "poly"
7 max_iter: 5000
8 power: 1
9 momentum: 0.9
10 weight_decay: 0.0005
11 snapshot: 50
12 snapshot_prefix: "/home/xianyu/programming/atlas200dk/code/output/resnet-18-reg"
13 random_seed: 0
14 test_initialization: false
15 solver_mode: CPU
16
终端 调试控制台 问题 输出
(base) xianyu@LAPTOP-V9JP5BP5:~/programming/atlas200dk$

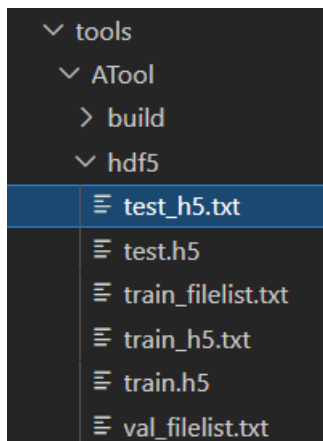
```

```

12 momentum: 0.9
13 momentum2: 0.999
14 # since Adam dynamically changes the learning rate, we set the base learning
15 # rate to a fixed value
16 lr_policy: "poly"
17 # Display every 100 iterations
18 display: 10
19 # The maximum number of iterations
20 max_iter: 5000
21 # snapshot intermediate results
22 snapshot: 50
23 snapshot_prefix: "/home/xianyu/programming/atlas200dk/code/output/resnet-18-reg-adam"
24 # solver mode: CPU or GPU
25 type: "Adam"
26 solver mode: CPU

```

还有这个文件夹下的test_h5.txt,train_h5.txt



```
code > tools > ATool > hdf5 > ≡ test_h5.txt
1  /home/xianyu/programming/atlas200dk/code/tools/ATool/hdf5/test.h5
2
code > tools > ATool > hdf5 > ≡ train_h5.txt
1  /home/xianyu/programming/atlas200dk/code/tools/ATool/hdf5/train.h5
2
```

到这里差不多就行了，切换到ResNet-18-Caffemodel-on-ImageNet目录下运行`train.sh`应该就可以了。

参考资料

[wsl基本命令](#)

[Windows与wsl\(Ubuntu\)的文件互访](#)

[在Ubuntu20.04上安装anaconda](#)